

2026

6 PRE-CODING DOCUMENTS

UK Local Grocery Store

Hyperlocal grocery delivery platform — complete pre-coding documentation covering requirements, architecture, flow, design, schema, and implementation plan

Project: UK STORE [DEMO]

Version: 2.0

Generated: June 2026

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Contents

01 PRD — Product Requirements Document

02 TRD — Technical Requirements Document

03 App Flow Document

04 UI/UX Design Brief

05 Backend Schema Document

06 Implementation Plan

Product Requirements Document

App Name and One-Line Idea

UK Local Grocery Store

A hyperlocal grocery delivery platform that enables UK customers to browse, order, and receive groceries delivered to their door in as fast as 15 minutes, with full UK legal compliance including VAT handling, allergen display, Challenge 25 age verification, and GDPR data protection.

Target Users and Problem Statement

USER TYPE	DESCRIPTION	KEY PAIN POINTS
Customers	UK residents ordering groceries online for home delivery	Slow delivery, inaccurate ETAs, hidden fees, missing allergen info, no age verification transparency
Store Owners	Independent grocery shop operators managing online orders	No integrated system for orders, inventory, staff, and compliance; manual processes lead to errors
Managers	Day-to-day operations staff overseeing picking and delivery	Order tracking across chat/calls, no real-time visibility, inventory mismatches
Pickers	Staff who collect and pack items from shelves	No structured packing workflow, quality assurance gaps, no QA checklist
Riders	Delivery personnel fulfilling last-mile delivery	No map integration, no ETA control, no age verification at door, poor

USER TYPE	DESCRIPTION	KEY PAIN POINTS
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communication with customer

Core Features with Priorities

FEATURE	PRIORITY	DESCRIPTION
Postcode Gate and Delivery Zone	Must	Validate customer postcode against delivery zones before allowing orders
Product Catalog with UK Compliance	Must	Products with unit pricing, VAT rates, allergens, Challenge 25 flags, use-by dates
Cart and Checkout	Must	Server-side cart validation, VAT breakdown, delivery fee, promo codes, age verification prompt
Stripe Payments with Webhooks	Must	3D Secure payments, webhook-verified order confirmation, idempotent processing
Order Management Kanban	Must	4-column board: New, Packing, Out for Delivery, Delivered with role-restricted views
Rider-Driven ETA and Live Tracking	Must	Rider sets manual ETA, customer sees live countdown, ETA adjustments (max 3)
Admin Dashboard	Must	Products, categories, staff, finance, promo codes, banners, system settings
RBAC Authentication	Must	5 roles: Customer, Owner, Manager, Picker, Rider with strict permission boundaries
UK Compliance Layer	Must	GDPR, CRA 2015, FIC 2014, Challenge 25, VAT Act 1994, PECR — 15 regulations covered
Customer Mobile App (CapacitorJS)	Should	Native wrapper for iOS/Android with push notifications, biometric login, offline cart
Notification System	Should	Email receipts (Resend), SMS delivery updates (Twilio), push notifications, in-app alerts

FEATURE	PRIORITY	DESCRIPTION
Product Recall Feature	Should	Admin-initiated recall that emails all affected customers within 7 days
Guest Checkout	Should	Order with email-only, prompt account creation after purchase
Wastage Logging	Could	Track expired/damaged stock for inventory management
Delivery Subscription Pass	Could	Monthly delivery pass for unlimited free delivery [VERIFY: Subscription model not yet confirmed by business]
AI Product Recommendations	Won't (v1)	Machine learning based product suggestions — deferred to v2

User Roles and Permissions

ROLE	AUTH METHOD	SESSION	ACCESS SCOPE
Customer	Magic link / biometric	7 days	Browse, order, track, reorder, manage account
Owner	Email + password + 2FA	30 min idle	Full dashboard: finance, settings, audit, staff salary, system config
Manager	Email + password + 2FA	30 min idle	Products, categories, banners, promos, staff (no system settings/salary)
Picker	Email + password	30 min idle	Packing queue only
Rider	Email + password	30 min idle	Delivery queue: map, ETA, customer phone, age verification

User Stories

- **As a customer**, I want to enter my postcode and instantly know if delivery is available, so that I don't waste time browsing unavailable services.
- **As a customer**, I want to see unit pricing (e.g., 1.40 GBP/100g) on every product, so that I can compare value fairly.
- **As a customer**, I want allergens highlighted in bold on product pages, so that I can quickly identify safety risks.
- **As a customer**, I want a live countdown timer when my order is out for delivery, so that I know exactly when to expect it.
- **As a rider**, I want to manually set my delivery ETA based on local knowledge, so that customers get accurate time estimates.
- **As a rider**, I want a "Cannot Verify" button for Challenge 25 items, so that age-restricted items are handled legally.
- **As a manager**, I want a Kanban board that flashes when new orders arrive, so that I can assign pickers and riders immediately.
- **As an owner**, I want VAT collected per month broken down by rate (0%, 5%, 20%) with CSV export, so that I can file MTD returns.
- **As a customer**, I want to delete my account and have my data anonymised, so that I can exercise my GDPR right to erasure.
- **As a picker**, I want a QA checklist on the packing screen, so that I can verify item quality before dispatch.

Success Metrics (KPIs)

<15

MIN AVG DELIVERY

>90

LIGHTHOUSE SCORE

<3s

TIME TO INTERACTIVE

>95%

ON-TIME DELIVERY

<0.75%

CHARGEBACK RATE

>4.5

APP STORE RATING

MVP Scope

IN Scope (v1)

- Postcode gate with delivery zone validation
- Product catalog with UK-compliant pricing
- Cart, checkout, Stripe payments
- Kanban order management
- Rider ETA and live customer tracking
- Admin dashboard (products, staff, finance)
- UK compliance (GDPR, VAT, allergens, Challenge 25)
- Customer mobile app (CapacitorJS)
- Email and SMS notifications
- RBAC with 5 roles

OUT of Scope (v1)

- AI-based product recommendations
- Delivery subscription pass
- Multi-store / franchise management
- Loyalty points system
- Real-time rider GPS tracking on map
- Recipe suggestions and meal kits
- Smart home integration
- Voice ordering
- Social media login (Google, Facebook)
- Internationalisation (non-UK markets)

Technical Requirements Document

Frontend Stack

LAYER	TECHNOLOGY	PURPOSE	COST
Framework	Next.js 14+ (App Router)	Storefront + Admin in Turborepo monorepo	Free
Language	TypeScript (strict)	Type safety, no any types	Free
Styling	Tailwind CSS	Responsive, utility-first, mobile-first	Free
State	Zustand + Server Components	Lightweight client state, server-first data	Free
Mobile	CapacitorJS 5	Native shell for iOS/Android (Customer App only)	Free

Backend Stack

LAYER	TECHNOLOGY	PURPOSE	COST
API	Next.js API Routes (Vercel Serverless)	REST endpoints, all business logic server-side	Free tier
Database	Turso (libSQL / SQLite)	Edge-replicated primary data store	Free tier
ORM	Drizzle ORM [VERIFY: TrishulHub standard is Prisma — this project uses Drizzle for Turso compatibility]	Type-safe queries, parameterised, migration support	Free

LAYER	TECHNOLOGY	PURPOSE	COST
Cache	Vercel KV (Redis)	Session storage, rate limiting, cart cache	Free tier
Auth	NextAuth.js v5	Magic link (customer), credentials + 2FA (staff)	Free
Payments	Stripe	Card payments, 3D Secure, webhooks, refunds	1.5% + 20p/tx
File Storage	Vercel Blob	Product images, receipts	Free tier
Email	Resend	Order confirmations, receipts, recall notices	Free (100/day)
SMS	Twilio	Delivery updates, ETA notifications	3-5p/UK SMS
Geocoding	Postcodes.io API	UK postcode validation and geolocation	Free
Monitoring	Sentry	Error tracking, performance monitoring	Free tier
CI/CD	GitHub Actions	Automated testing and deployment	Free tier

Database

Primary: Turso (libSQL) — edge-replicated SQLite for fast reads from any UK location. UUID v4 primary keys. Full-text search via SQLite FTS5.

Cache Layer: Vercel KV (Redis) — session storage, rate limiting counters, cart caching for guest users.

Backups: Automated daily via Turso scheduled backups, encrypted, stored in separate region.

Retention: Order data 7 years (HMRC), PII deleted 30 days after account closure.

Authentication Method

USER TYPE	METHOD	SESSION STORAGE	EXPIRY
Customer (Web)	Magic link email	HTTP-only cookie	7 days
Customer (App)	Biometric (Face ID / fingerprint) + magic link fallback	HTTP-only cookie	7 days
Staff (All roles)	Email + password + optional TOTP 2FA	HTTP-only cookie	30 min idle

APIs Needed

ENDPOINT	METHOD	AUTH	MIN ROLE	DESCRIPTION
/api/products	GET	No	—	List/search products with pagination
/api/products/:id	GET	No	—	Product detail with allergens and ingredients
/api/categories	GET	No	—	All active categories with product counts
/api/auth/login	POST	No	—	Magic link or credential login

ENDPOINT	METHOD	AUTH	MIN ROLE	DESCRIPTION
/api/auth/register	POST	No	—	Customer registration
/api/delivery/validate-postcode	POST	No	—	Check if postcode is in delivery zone
/api/orders	POST	Cookie	Customer	Create order (server calculates total)
/api/orders/:id	GET	Cookie	Customer	Order status and details
/api/orders/:id/tracking	GET	Cookie	Customer	Live tracking data (status, ETA, rider info)
/api/payments/create-intent	POST	Cookie	Customer	Create Stripe PaymentIntent server-side
/api/payments/webhook	POST	Stripe Sig	—	Handle Stripe events (payment_success, refund)
/api/admin/orders	GET	Cookie	Picker	Order list with status filters
/api/admin/orders/:id/status	PATCH	Cookie	Picker	Update order status (Kanban)
/api/admin/orders/:id/assign-rider	PATCH	Cookie	Manager	Assign rider, send push notification
/api/admin/orders/:id/set-eta	PATCH	Cookie	Rider	Rider sets ETA, dispatches order
/api/admin/orders/:id/adjust-eta	PATCH	Cookie	Rider	Rider adjusts ETA (max 3 times)

ENDPOINT	METHOD	AUTH	MIN ROLE	DESCRIPTION
/api/admin/orders/:id/verify-age	PATCH	Cookie	Rider	Confirm age verification at delivery
/api/admin/orders/:id/deliver	PATCH	Cookie	Rider	Mark order as delivered
/api/admin/products	GET/POST	Cookie	Manager	List or create products
/api/admin/products/:id	PATCH	Cookie	Manager	Update product details and stock
/api/admin/finance/summary	GET	Cookie	Owner	Revenue, profit, VAT collected
/api/admin/settings	GET/PUT	Cookie	Owner	Store settings (status, delivery zones)

Cloud / Deployment Setup

- **Hosting:** Vercel (Hobby tier free, Pro at 20 GBP/month)
- **Database:** Turso (Starter free, Scaler at 29 GBP/month)
- **Domain:** UK domain (e.g., towndelivery.co.uk) via 123-Reg
- **SSL:** Included with Vercel hosting
- **CI/CD:** GitHub Actions + Vercel auto-deploy from main branch
- **Environments:** Preview (every PR), Staging (staging branch), Production (main branch)

Security Requirements

REQUIREMENT	IMPLEMENTATION
HTTPS everywhere	Vercel default, HSTS header
SQL injection prevention	Drizzle ORM parameterised queries, never raw SQL
XSS prevention	Content Security Policy header, no inline scripts
CSRF protection	SameSite=Strict cookies, CSRF token for state-changing requests
Rate limiting	100 req/min per IP (public), 300 req/min per user (authenticated)
Input validation	Zod schemas for all request bodies
PII encryption at rest	AES-256 for names, addresses, phone numbers
Payment card data	Never touches our server — Stripe Elements handles entirely
CORS	Only own domain, no wildcard Access-Control-Allow-Origin
Security headers	Strict-Transport-Security, X-Content-Type-Options, X-Frame-Options
Webhook signature verification	Stripe webhook secret, cryptographic verification on every event

Performance Requirements

<1.5s

FIRST CONTENTFUL
PAINT

<3s

TIME TO INTERACTIVE
(4G)

>90

LIGHTHOUSE SCORE

<100ms

CART OPERATIONS

<2s

KANBAN LOAD TIME

<5s

WEBSOCKET
RECONNECT

Third-Party Integrations

SERVICE	PURPOSE	INTEGRATION METHOD
Stripe	Payments, refunds, webhooks	Server-side SDK + webhook endpoint
Postcodes.io	UK postcode validation and geolocation	REST API (free, no auth)
Resend	Transactional and marketing emails	Server-side SDK
Twilio	SMS delivery updates	Server-side SDK + STOP keyword handling
Vercel KV	Session storage, rate limiting	Vercel SDK (@vercel/kv)
Vercel Blob	Product image storage	Vercel SDK (@vercel/blob)
Sentry	Error and performance monitoring	Next.js SDK (@sentry/nextjs)
Google Maps / Leaflet.js	Rider map view, customer pin	iframe embed + "Open in Maps" link
CapacitorJS	Native device features (Customer App)	Plugins for geolocation, push, biometric

App Flow Document

All Pages / Screens

SCREEN	USER TYPE	KEY COMPONENTS
Postcode Entry Popup	Customer	GPS detect, manual input, delivery zone check, unavailable area message + email signup
Homepage	Customer	Sticky header (postcode, timer, search, cart), banner carousel, category sliders, product grids
Category Listing	Customer	Left sidebar sub-categories, right product grid (6×3), instant sub-category switching
Product Detail	Customer	Image gallery, name, brand, weight, price, unit price, VAT badge, allergens (bold), ingredients, use-by date, Challenge 25 badge, Add to Cart, cross-selling rows
Cart Sidebar	Customer	Slide-out, quantity controls, running bill, promo code, delivery fee, free shipping goal, Proceed to Pay
Checkout	Customer	Address confirm, order summary, VAT breakdown, cancellation rights, age verification checkbox, Stripe payment
Order Tracking	Customer	Live countdown timer, status timeline, rider contact card, Call Rider button
Order History	Customer	Past orders list, one-tap reorder, out-of-stock warnings
Account Settings	Customer	Biometric login, saved addresses, notification preferences, Delete My Account (GDPR)

SCREEN	USER TYPE	KEY COMPONENTS
Admin Dashboard	Owner / Manager	Metric cards (revenue, profit, orders), date filter
Kanban Board	All Staff	4 columns: New (flashing), Packing, Out for Delivery, Delivered — role-restricted views
Product Management	Manager	Add/edit products, image upload, VAT selector, Challenge 25 toggle, allergens, stock management, kill switch
Rider Delivery Screen	Rider	Delivery queue, order detail, map view, ETA presets (15/30/45/60), Start Delivery, +10 min adjustment, Verify Age / Cannot Verify, Mark Delivered
Finance Dashboard	Owner	Revenue, profit, VAT collected, product cost, CSV export for MTD
Staff Management	Manager	Employee roster, attendance toggle, order counter, salary tracker
System Settings	Owner	Store open/close switch, delivery areas, free delivery threshold, Stripe keys

Customer Journey — End to End

- 1 Postcode Gate** — Customer enters or GPS-detects postcode. Validated against delivery_zones via Postcodes.io. Invalid → "Sorry, we don't deliver to your area" + email signup.
- 2 Browse Homepage** — Banner carousel, category sliders, product cards with unit pricing, discount badges, 18+ badges.
- 3 Search or Navigate** — Live predictive search or category sidebar navigation. Products filtered by sub-category instantly.
- 4 View Product** — Full detail: images, allergens (bold), ingredients, use-by date, unit price, Challenge 25 warning. Add to Cart.
- 5 Review Cart** — Slide-out sidebar. Quantity controls, bill summary, promo code, delivery fee, age-restricted item warning.

- 6 **Checkout** — Address confirm, order summary, VAT breakdown, cancellation rights, 18+ confirmation checkbox. Proceed to Pay.
- 7 **Payment** — Stripe 3D Secure modal. Server creates PaymentIntent, verifies total from DB (not client). Webhook confirms.
- 8 **Order Confirmed** — Email receipt via Resend, order appears in Kanban "New" column (flashing + sound alert).
- 9 **Packing** — Picker accepts order, follows QA checklist, marks as Packed.
- 10 **Dispatch** — Manager assigns Rider. Rider reviews address + map, sets manual ETA (15/30/45/60 min), taps Start Delivery. Customer receives SMS + push with countdown.
- 11 **Live Tracking** — Customer sees countdown timer, status timeline, rider info. Rider can adjust ETA (+10 min, max 3 times).
- 12 **Delivery** — Challenge 25 check at door. Rider marks Delivered. Confirmation email + SMS to customer. Feedback prompt.

Navigation Flow

Customer (Web/App): Postcode Gate → Homepage → Category Page → Product Detail → Cart Sidebar → Checkout → Payment → Order Tracking → Order History

Staff (Responsive Web): Login → Dashboard → Kanban Board (role-filtered) → Order Detail → Action (Pack/Assign/Set ETA/Deliver)

Admin (Desktop): Login → Dashboard → Products | Staff | Finance | Settings (RBAC-controlled)

Error and Edge States

STATE	TRIGGER	USER EXPERIENCE
Empty category	No products in category	Friendly illustration + "No items yet" message + suggest alternative categories

STATE	TRIGGER	USER EXPERIENCE
Out of stock	stock_count = 0	Badge on product card, Add button disabled, polling every 30s
Payment failed	Card declined / Stripe error	Clear message + alternative payment prompt, cart preserved
Delivery unavailable	Postcode outside zone	Modal with explanation + email signup for future
Offline	No internet connection	Service Worker cached content, cart in localStorage, "Reconnecting..." indicator
Server error (5xx)	Server crash / DB timeout	Full-page error with retry button, Sentry alert, exponential backoff
ETA expired	Delivery past estimated time	Customer sees "Running slightly late", Manager alerted, SMS sent to customer
Age verification failed	Rider cannot verify age at door	Restricted items removed, partial refund initiated, customer notified

Login / Signup Flow

Customer: Enter email → magic link sent → click link → authenticated (7-day session). Guest checkout available with email-only. Post-purchase: prompt to set password for full account.

Staff: Enter email + password → optional TOTP 2FA → authenticated (30-min idle session). All sessions via HTTP-only cookies.

Mobile App: First launch → location permission → GPS postcode detection → biometric login (Face ID / fingerprint) → fallback to magic link.

Payment Flow

Customer clicks "Proceed to Pay" → Server creates Stripe PaymentIntent with amount calculated from DB (not client) → Stripe 3D Secure modal appears →

Customer authenticates → Stripe sends webhook to /api/payments/webhook
→ Server verifies webhook signature → Marks order "Paid" → Decrements
inventory → Sends email receipt → Kanban "New" column flashes.

Failure path: Payment fails → Order marked "Abandoned" → Inventory
restored → Customer receives retry email → Cart preserved 15 minutes.

UI/UX Design Brief

Design Style and Mood

Style: Clean, modern, trust-building grocery retail. Minimalist with bold accent colours for calls-to-action. Mobile-first with thumb-friendly interaction zones.

Mood: Fast, reliable, fresh. The design should communicate speed (delivery timer, quick-add buttons) and trust (compliance badges, secure payment indicators, clear pricing).

Philosophy: Search-first, not browse-first. Most grocery shoppers know what they want — the search bar is the hero. Categories and sliders support discovery for undecided shoppers.

Color Palette

Primary



Green family — trust, freshness, "go" action

Accent / CTA



Rose = primary CTA, Orange = urgency/savings, Purple = Challenge 25 / age-restricted

ROLE	HEX	USAGE
Primary Green	#10b981	Add buttons, success states, delivery timer, in-stock badges

ROLE	HEX	USAGE
Primary Dark	#059669	Hover states, active navigation
CTA Rose	#f43f5e	"Proceed to Pay" button, urgency indicators, Kanban flash
Accent Orange	#fb923c	Discount badges, savings display, "Low Stock" warning
Accent Purple	#a855f7	Challenge 25 badge (18+), age-restricted indicators
Background Light	#fafafa	Page backgrounds (storefront)
Background Card	#ffffff	Product cards, modals, sidebar
Text Primary	#111827	Headings, product names, prices
Text Secondary	#6b7280	Descriptions, labels, metadata
Text Muted	#9ca3af	Timestamps, footnotes, disabled state
Border	#e5e7eb	Card borders, dividers, input borders
Error Red	#ef4444	Form errors, out-of-stock, "Cannot Verify"
Admin Dark	#0f172a	Admin dashboard background
Admin Card	#1e293b	Admin dashboard cards

Typography

ELEMENT	FONT	SIZE	WEIGHT	LINE HEIGHT
Hero / Delivery Timer	Inter	48-64px	800-900	1.1
Page Title (H1)	Inter	28-36px	700	1.2
Section Title (H2)	Inter	20-24px	700	1.3
Card Title (H3)	Inter	16-18px	600	1.4

ELEMENT	FONT	SIZE	WEIGHT	LINE HEIGHT
Body Text	Inter	14-16px	400	1.6
Product Price	Inter	18-20px	700	1.2
Unit Price	Inter	12-13px	400	1.3
Badge / Tag	Inter	10-12px	600	1.2
Button	Inter	14-16px	600	1.0
Code / Monospace	JetBrains Mono	13-14px	400	1.5

Component Style Guide

COMPONENT	STYLE
Product Card	White background, 8px radius, 1px border, product image top, name + weight, bold price, unit price below, green "ADD" button overlay (48×48px min), discount badge top-right
Add to Cart Button	Green (#10b981), white text, 16px height, 8px radius, bold, full-width on mobile, bounce animation on add
Proceed to Pay Button	Rose (#f43f5e), white text, 56px height, 12px radius, bold, fixed bottom on mobile, massive tap target
Category Slider	Horizontal scroll, bold category name left, "See All" link right, product cards in scroll row, snap scrolling
Cart Sidebar	Slide from right, overlay backdrop, green savings bar top, product rows with +/- counters, bill summary, promo code input
Kanban Card	White/light card, order ID, customer name, item count, total, status badge, flash animation on "New" cards
ETA Preset Buttons	4 large buttons (15/30/45/60 min), 48×48px min, selected state highlighted green, one-tap selection
18+ Badge	Purple (#a855f7) pill badge, white "18+" text, on product cards and detail pages

COMPONENT	STYLE
Discount Badge	Orange (#fb923c) pill badge, white "22% OFF" text, top-right of product card
Countdown Timer	Large (48px+), bold, green text, "Your order arrives in XX min", changes to "Almost there!" under 5 min

Layout Rules and Grid System

BREAKPOINT	WIDTH	LAYOUT
Mobile S	320-374px	Single column, bottom sheet cart, hamburger nav
Mobile	375-428px	Single column, sticky add button, swipe cards
Tablet	429-768px	Two-column product grid, sidebar cart
Desktop S	769-1024px	Three-column grid, expanded nav, sidebar categories
Desktop	1025-1440px	Four-column grid, full navigation bar
Desktop XL	1441px+	Six-column grid, wider spacing

Mobile / Desktop Behavior

Mobile-first: All designs start from 375px and scale up. Thumb zones prioritised — CTAs in bottom 60% of screen. Minimum tap target 44×44px (WCAG). Swipe gestures for cart and categories.

Desktop: Expanded navigation, multi-column grids, sidebar cart becomes inline. Admin dashboard fully featured on desktop with data tables and charts.

Rider mobile web: Optimised for one-handed use on phone browsers. Large touch targets (48×48px min), high contrast, tel: links for calling, embedded map view.

Overall UX Principles

- **Speed perceived:** Delivery timer, skeleton loading (200ms), optimistic UI updates for cart
- **Trust signals:** Stripe badge, VAT breakdown, allergen emphasis, 30-day guarantee statement, ICO registration
- **Compliance visible:** Unit pricing on every card, Challenge 25 badges, cookie consent, cancellation rights at checkout
- **Glanceable information:** Countdown timer, order status timeline, low stock badges — no scrolling for critical info
- **Forgiving design:** Cart preserved on payment failure, 15-min retry window, guest checkout, ETA adjustment grace

Backend Schema Document

Database: Turso (libSQL / SQLite)

All tables use UUID v4 primary keys. Foreign keys enforced. Timestamps stored as ISO 8601 TEXT. Order items snapshot price and VAT rate at time of purchase for historical accuracy.

Core Tables

users

COLUMN	TYPE	CONSTRAINTS	PURPOSE
id	TEXT	PRIMARY KEY	UUID v4 unique identifier
email	TEXT	UNIQUE, NOT NULL	Login and communication
phone	TEXT		Delivery contact number
full_name	TEXT	NOT NULL	Customer display name
password_hash	TEXT		bcrypt hash (staff only)
role	TEXT	DEFAULT "customer"	customer / owner / manager / picker / rider
postcode	TEXT		Default delivery postcode

COLUMN	TYPE	CONSTRAINTS	PURPOSE
is_banned	BOOLEAN	DEFAULT FALSE	Fraud prevention flag
email_marketing_consent	BOOLEAN	DEFAULT FALSE	PECR marketing consent
sms_marketing_consent	BOOLEAN	DEFAULT FALSE	PECR SMS consent (always opt-in)
marketing_consent_date	TEXT		When consent was given
created_at	TEXT	DEFAULT CURRENT_TIMESTAMP	Account creation date
updated_at	TEXT		Last profile update

products

COLUMN	TYPE	CONSTRAINTS	PURPOSE
id	TEXT	PRIMARY KEY	UUID v4
name	TEXT	NOT NULL	Product display name
description	TEXT		Full product description
category_id	TEXT	FK → categories.id	Product category
brand_id	TEXT	FK → brands.id	Product brand
price	REAL	NOT NULL	Selling price in GBP (VAT-inclusive)
compare_at_price	REAL		Original price (for discount display)

COLUMN	TYPE	CONSTRAINTS	PURPOSE
unit_price	TEXT	NOT NULL	e.g. "1.20/100g" (UK law requirement)
vat_rate	REAL	NOT NULL	0.0, 0.05, or 0.20
weight_value	REAL		Numeric weight or volume
weight_unit	TEXT		g, kg, ml, L
image_url	TEXT		Product photograph URL (Vercel Blob)
stock_count	INTEGER	DEFAULT 0	Current inventory level
low_stock_threshold	INTEGER	DEFAULT 10	Alert trigger level
is_active	BOOLEAN	DEFAULT TRUE	Kill switch for stock-out
challenge_25	BOOLEAN	DEFAULT FALSE	Age-restricted item flag
allergens	TEXT		Comma-separated allergen list
ingredients	TEXT		Full ingredients list
is_perishable	BOOLEAN	DEFAULT FALSE	Perishable goods flag (cancellation rules)
preparation_type	TEXT		prepacked / ppds / loose (Natasha's Law)
use_by_date	TEXT		For fresh items
created_at	TEXT	DEFAULT CURRENT_TIMESTAMP	Product creation date

orders

COLUMN	TYPE	CONSTRAINTS	PURPOSE
id	TEXT	PRIMARY KEY	UUID v4
user_id	TEXT	FK → users.id	Customer reference
status	TEXT	NOT NULL	new / packing / out_for_delivery / delivered / cancelled
subtotal	REAL	NOT NULL	Pre-VAT item total
vat_total	REAL	NOT NULL	Total VAT collected
delivery_fee	REAL	DEFAULT 0	Delivery charge
discount_amount	REAL	DEFAULT 0	Promo code discount
total	REAL	NOT NULL	Final amount charged
stripe_payment_intent_id	TEXT	UNIQUE	Stripe payment reference
delivery_address	TEXT	NOT NULL	Full delivery address
delivery_postcode	TEXT	NOT NULL	UK postcode
customer_phone	TEXT		Contact number for delivery
assigned_picker_id	TEXT	FK → users.id	Staff packing the order
assigned_rider_id	TEXT	FK → users.id	Staff delivering the order

COLUMN	TYPE	CONSTRAINTS	PURPOSE
estimated_delivery_minutes	INTEGER		ETA set manually by rider
estimated_delivery_at	TEXT		Calculated: dispatched_at + ETA minutes
dispatched_at	TEXT		When rider starts delivery
delivered_at	TEXT		When rider confirms delivery
eta_adjustment_count	INTEGER	DEFAULT 0	Number of ETA adjustments (max 3)
age_verified	BOOLEAN	DEFAULT FALSE	Challenge 25 check passed
promo_code_id	TEXT	FK → promo_codes.id	Applied discount code
cancellation_reason	TEXT		If cancelled, record reason
age_verification_confirmed	BOOLEAN	DEFAULT FALSE	Customer confirmed 18+ at checkout
created_at	TEXT	DEFAULT CURRENT_TIMESTAMP	Order timestamp
updated_at	TEXT		Last status change

order_items

COLUMN	TYPE	CONSTRAINTS	PURPOSE
id	TEXT	PRIMARY KEY	UUID v4
order_id	TEXT	FK → orders.id	Parent order
product_id	TEXT	FK → products.id	Product at time of order
quantity	INTEGER	NOT NULL	Number of items
unit_price	REAL	NOT NULL	Price per item (snapshot)
vat_rate	REAL	NOT NULL	VAT rate at time of order (snapshot)
line_total	REAL	NOT NULL	unit_price * quantity

Supporting Tables

TABLE	KEY COLUMNS	PURPOSE
categories	id, name, slug, display_order, is_active	Product categorisation (e.g., Dairy, Meat, Bakery)
brands	id, name, slug, logo_url	Brand filtering and display
promo_codes	id, code, discount_type, discount_value, min_order, max_uses, used_count, expires_at	Marketing promotions
delivery_zones	id, postcode_prefix, max_distance_km, is_active	Delivery area management
banners	id, image_url, link_target, display_order, is_active	Homepage banner carousel
staff_attendance	id, user_id, date, status	Daily attendance tracking
wastage_log	id, product_id, quantity, reason, logged_at	Expired/damaged stock tracking

TABLE	KEY COLUMNS	PURPOSE
audit_log	id, user_id, action, entity, entity_id, timestamp	Security and compliance audit trail
order_tracking	id, order_id, status, eta_minutes, rider_lat, rider_lng, timestamp	Real-time tracking event log
notifications	id, user_id, type, message, is_read, created_at	In-app and push notification queue
delivery_temperature_log	id, order_id, temperature_reading, checked_at	Food safety temperature records

Indexes

TABLE	INDEX	COLUMNS	PURPOSE
products	idx_products_category	category_id	Category page queries
products	idx_products_active	is_active	Active product filtering
products	idx_products_use_by	use_by_date	Daily expiry cron job
orders	idx_orders_user	user_id	Customer order history
orders	idx_orders_status	status	Kanban board queries
orders	idx_orders_stripe	stripe_payment_intent_id	Webhook lookup
order_items	idx_oi_order	order_id	Order detail queries

TABLE	INDEX	COLUMNS	PURPOSE
order_tracking	idx_ot_order	order_id	Tracking event lookup
audit_log	idx_audit_entity	entity, entity_id	Entity-specific audit queries
notifications	idx_notif_user_unread	user_id, is_read	Unread notification count

Foreign Key Relationships

- `products.category_id` → `categories.id`
- `products.brand_id` → `brands.id`
- `orders.user_id` → `users.id`
- `orders.assigned_picker_id` → `users.id`
- `orders.assigned_rider_id` → `users.id`
- `orders.promo_code_id` → `promo_codes.id`
- `order_items.order_id` → `orders.id` (CASCADE DELETE)
- `order_items.product_id` → `products.id`
- `order_tracking.order_id` → `orders.id` (CASCADE DELETE)
- `notifications.user_id` → `users.id`
- `audit_log.user_id` → `users.id`

Data Ownership Rules

- Customers can only access their own orders, profile, and addresses
- Pickers can only see orders in "New" and "Packing" status
- Riders can only see orders assigned to them in "Out for Delivery"
- Managers can see all orders and modify products, categories, banners, promo codes, and staff

- Owners have unrestricted access to all data including finance and audit logs
- All data modifications are logged in audit_log with user_id, action, timestamp, and before/after values

Authentication / Session Handling

Session Storage: Vercel KV (Redis) via HTTP-only cookies. Session ID is a cryptographically random token.

Customer Sessions: 7-day expiry, no idle timeout. Extend on each request.

Staff Sessions: 30-minute idle timeout. 2FA optional (TOTP). Force re-auth on role elevation.

Guest Checkout: Cart stored in Vercel KV with session ID. No user account required. Post-purchase prompt to create account.

Token Storage: Stripe tokens never stored in our database. PaymentIntent IDs stored for reconciliation only.

Implementation Plan

8-Week Development Timeline

Week 1	Foundation — Monorepo, DB migrations, auth skeleton, account setup, ICO registration
Week 2-3	Core Storefront — Postcode gate, products, cart, checkout, payment integration
Week 4	Admin Dashboard — Product CRUD, Kanban board, staff management, finance overview
Week 5	Payments and Webhooks — Stripe webhook handler, idempotent processing, refund workflow
Week 6	Compliance Layer — UK GDPR, VAT, privacy policy, WCAG, cookie consent, allergen display
Week 7	Notifications — Email receipts, SMS delivery updates, push notifications, admin alerts
Week 8	Testing and QA — E2E tests, security audit, accessibility audit, load testing, UAT
Week 9+	Launch — Live Stripe, app store submission, monitoring, first 20 orders monitored

Phase 1: Foundation (Week 1)

Day 1-2: Account Setup

- Register UK domain name (e.g., towndelivery.co.uk)
- Create GitHub organisation with branch protection rules
- Set up Vercel project with preview deployments
- Create Turso database and obtain connection string

- Register with ICO as a data controller (40 GBP/year)
- Create Stripe account in UK mode

Day 3-4: Project Initialisation

- Initialise Turborepo monorepo with apps/web and apps/admin
- Configure shared packages: UI components, database client, types
- Set up Drizzle ORM with Turso driver; write initial migration for all tables
- Configure NextAuth.js v5 with credentials provider and magic link
- Set up Vercel KV for session storage and rate limiting

Day 5: Auth and RBAC

- Run database migrations and seed with sample product data
- Implement user registration and login API routes
- Implement RBAC middleware for admin route protection
- Write API integration tests for auth endpoints
- Deploy to Vercel preview and verify all services connect

Phase 2: Core Storefront (Weeks 2-3)

Week 2: Browsing and Discovery

- Implement postcode gate with Postcodes.io API validation
- Build sticky header with search bar, location picker, cart widget
- Create category slider and product card components with unit pricing
- Build homepage banner carousel with admin-configurable slides
- Implement category listing page with left sidebar and product grid
- Add Challenge 25 badge rendering for age-restricted products
- Implement allergen information display on product cards and detail pages

Week 3: Cart, Checkout, Payment

- Build slide-out cart sidebar with quantity controls and bill summary
- Implement server-side cart validation (stock check, price verification)
- Build checkout form with address auto-complete using postcode lookup
- Display VAT breakdown on receipt (0%, 5%, 20% line items)
- Implement Stripe PaymentIntent creation on the server

- Build order confirmation page with estimated delivery time
- Add guest checkout flow with email-only identification

Phase 3: Admin Dashboard (Week 4)

- Build real-time Kanban board with WebSocket updates for order fulfilment
- Implement product CRUD with image upload, VAT selector, Challenge 25 toggle, allergen fields
- Build stock management with kill switch and low stock alerts
- Implement finance overview: revenue, profit, VAT collected, CSV export for MTD
- Build employee management with attendance tracking and daily performance
- Implement category and brand management interfaces
- Add audit log viewer for Owner role
- Build promo code management with usage restrictions and expiry dates

Phase 4: Payments and Webhooks (Week 5)

- Implement Stripe webhook handler with signature verification
- Handle `payment_intent.succeeded`: mark order "Paid", decrement inventory
- Handle `payment_intent.failed`: mark order "Abandoned", restore inventory
- Handle `charge.refunded`: update order status, log refund reason
- Implement idempotent webhook processing (prevent duplicate updates)
- Build refund initiation from admin panel (Manager+ role only)
- Set up Stripe test mode and create test scenario scripts
- Implement receipt email sending via Resend on successful payment

Phase 5: Compliance Layer (Week 6)

- Implement cookie consent banner with granular preference controls
- Create privacy policy page and terms and conditions page
- Build cancellation and returns information page
- Implement VAT receipt generation with HMRC-compliant format
- Verify unit pricing display on all product pages
- Ensure Challenge 25 age verification prompt at checkout
- Conduct WCAG 2.1 AA accessibility audit

- Add allergen filtering and prominent allergen display
- Implement data export for Subject Access Requests
- Build "Delete My Account" GDPR erasure workflow

Phase 6: Notifications (Week 7)

- Configure Resend for transactional emails (order confirmation, receipt, recall)
- Configure Twilio for SMS delivery updates with tracking link
- Implement push notifications for customer order status changes
- Build admin sound alert for new orders on Kanban board
- Implement rider dispatch notification via WebSocket + browser notification
- Add order status change notifications to customer
- Implement delivery ETA SMS with tracking link

Phase 7: Testing and QA (Week 8)

- Write end-to-end tests with Playwright for all critical user journeys
- Verify VAT calculations: 0% on fresh food, 20% on alcohol and soft drinks
- Test Challenge 25 flow: age-restricted items prompt verification
- Test RBAC: Picker cannot access Finance; Rider cannot modify products
- Load test with k6: simulate 100 concurrent users
- Security scan with OWASP ZAP: SQL injection, XSS, CSRF
- Accessibility audit with axe-core and manual screen reader testing
- Mobile device testing on iOS Safari and Android Chrome
- Stripe webhook testing with all event types
- User acceptance testing with store owner and staff

Phase 8: Launch (Week 9+)

- Switch Stripe from test mode to live mode with real bank account
- Deploy production build to Vercel and verify all services
- Submit Android app to Google Play Console (Physical Goods exemption)
- Submit iOS app to App Store Connect (Physical Goods exemption)
- Set up Sentry error monitoring with Slack/email alerts
- Configure Vercel Analytics for Core Web Vitals monitoring

- Monitor first 20 orders closely and resolve issues immediately

Cost Summary

ITEM	COST	NOTES
UK domain	10-15 GBP/year	e.g. towndelivery.co.uk
ICO registration	40 GBP/year	Data controller fee
Google Play Developer	25 GBP (one-time)	Android app publishing
Apple Developer	99 GBP/year	iOS app publishing
SSL certificate	0 GBP	Included with Vercel
One-time setup (Year 1)	174-184 GBP	
Monthly (Free Tier)	0 GBP/month	Sufficient for 100-200 orders/month
Monthly (Pro Tier)	~113 GBP/month	Needed for 500+ orders/month

Items Requiring Verification

The following assumptions were made during document generation. Please confirm or correct each item:

1. [VERIFY: ORM is Drizzle (for Turso compatibility) — TrishulHub standard is Prisma/PostgreSQL. Is this deviation approved?]
2. [VERIFY: Database is Turso (libSQL/SQLite) — TrishulHub standard is PostgreSQL. Is this deviation approved?]
3. [VERIFY: Monorepo tool is Turborepo — is this preferred over a single Next.js app?]
4. [VERIFY: Delivery subscription pass model — business has not confirmed if this will be offered]
5. [VERIFY: Map provider — Google Maps vs Leaflet.js not yet decided. Google Maps requires API key and billing.]

6. [VERIFY: Rider GPS tracking — blueprint specifies manual ETA only. Real-time GPS tracking on map is out of scope for v1. Confirmed?]
7. [VERIFY: Social login (Google/Facebook) — explicitly out of scope for v1. Is this acceptable?]
8. [VERIFY: Store domain name — towndelivery.co.uk used as placeholder. Actual domain TBD.]